

Exp No:4b

Data Inspection, Filtering Statistical analysis

Date:

AIM

To visualize and analyze the social media dataset using bar charts and scatter plots in Python, in order to compare platform-wise engagement and identify relationships between social media metrics such as Likes, Comments, Shares, and Views.

ALGORITHM

- Import Pandas, Matplotlib, and Seaborn libraries.
- Load the social media dataset into a DataFrame.
- Group and analyze the data based on Platform and Sentiment.
- Plot bar charts to compare platform-wise engagement and sentiment distribution.
- Plot scatter charts to visualize relationships between Likes, Views, Comments, and Shares, and display the graphs.

CODE:

1.

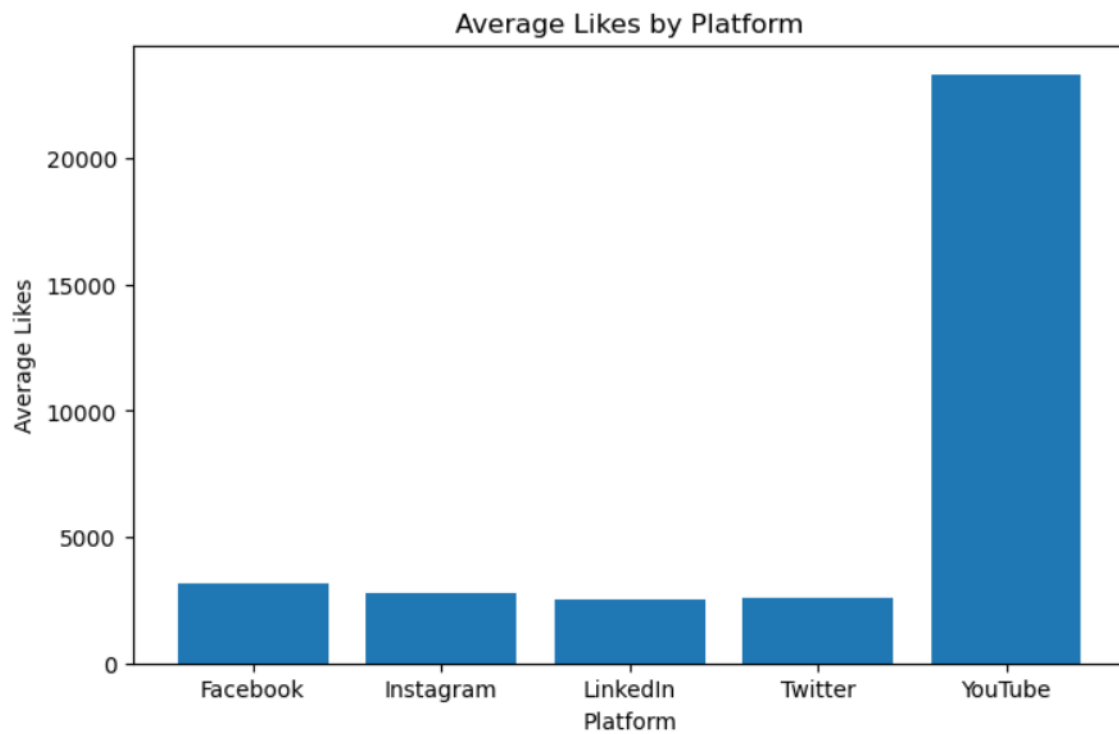
```
import matplotlib.pyplot as plt
import seaborn as sns

platform_likes = df.groupby("Platform")["Likes"].mean()

plt.figure(figsize=(8, 5))
plt.bar(platform_likes.index, platform_likes.values)

plt.xlabel("Platform")
plt.ylabel("Average Likes")
plt.title("Average Likes by Platform")

plt.show()
```



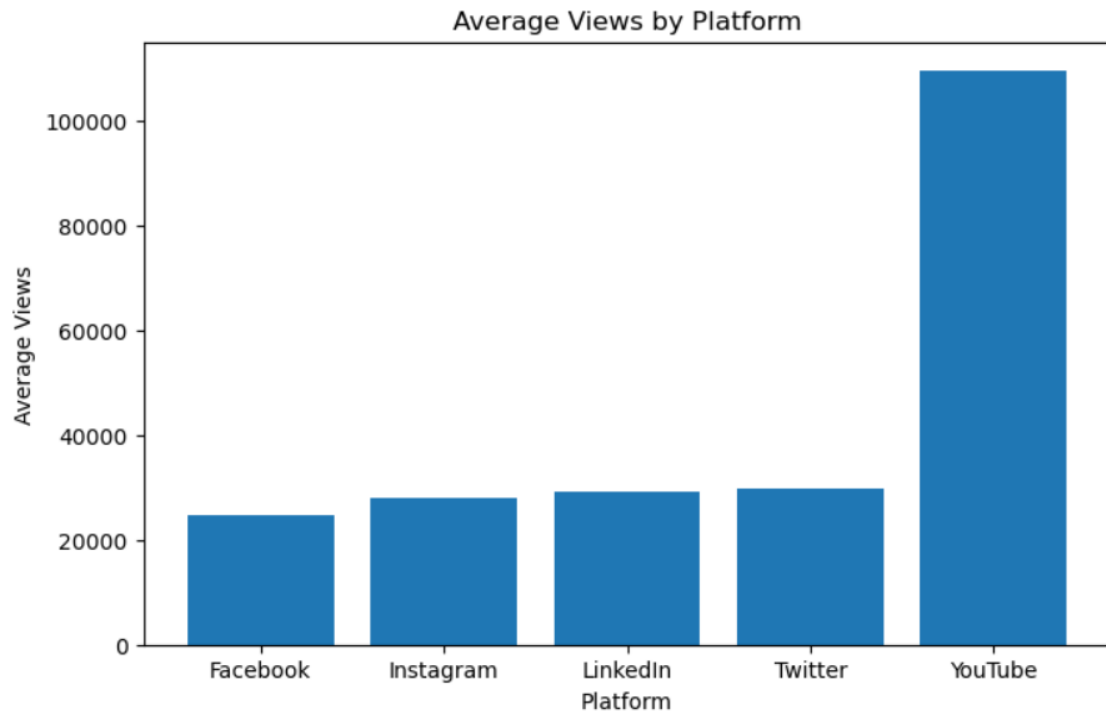
2.

```
platform_views = df.groupby("Platform")["Views"].mean()

plt.figure(figsize=(8, 5))
plt.bar(platform_views.index, platform_views.values)

plt.xlabel("Platform")
plt.ylabel("Average Views")
plt.title("Average Views by Platform")

plt.show()
```



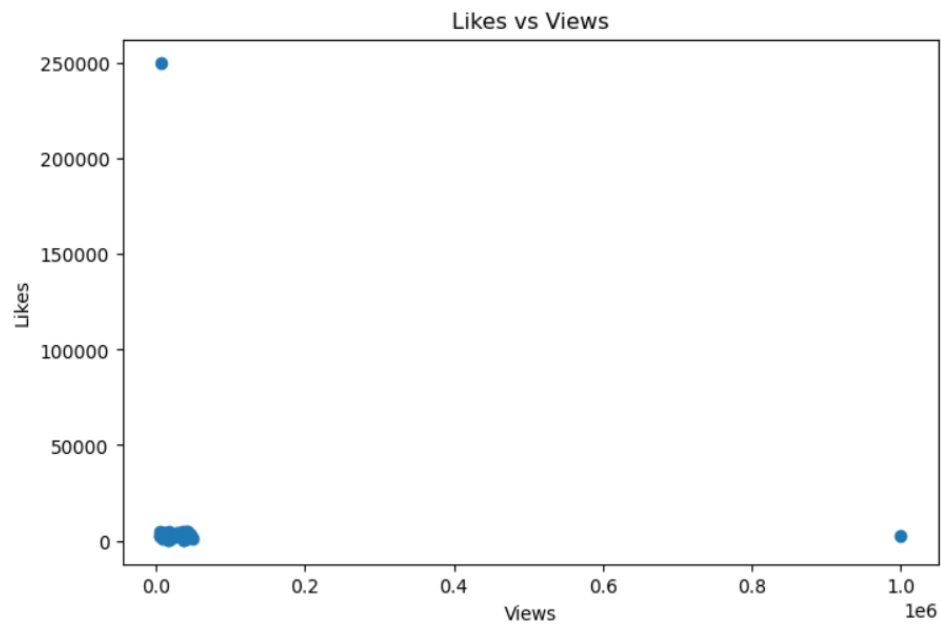
3.

```
plt.figure(figsize=(8, 5))

plt.scatter(
    df["Views"],
    df["Likes"]
)

plt.xlabel("Views")
plt.ylabel("Likes")
plt.title("Likes vs Views")

plt.show()
```



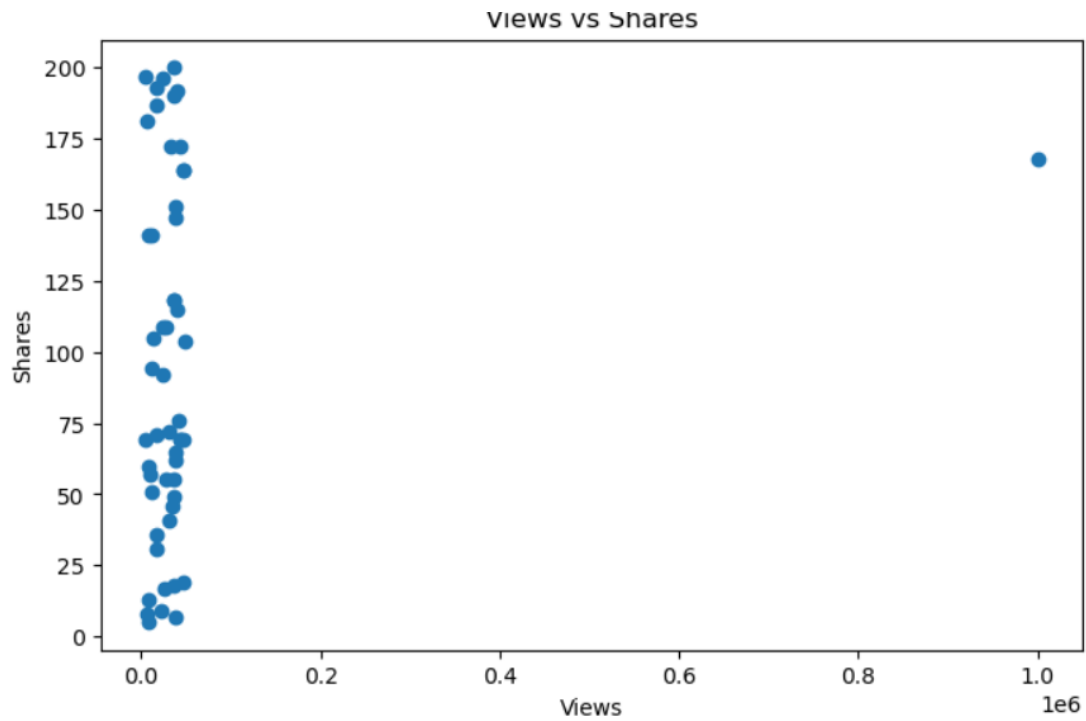
4.

```
plt.figure(figsize=(8, 5))

plt.scatter(
    df["Views"],
    df["Shares"]
)

plt.xlabel("Views")
plt.ylabel("Shares")
plt.title("Views vs Shares")
|
plt.show()
```





RESULT: Thus, the social media dataset was successfully analyzed using bar and scatter plots, providing a visual understanding of platform-wise engagement, sentiment distribution, and relationships between social media metrics.